



第三单元 热学

Unit 3 Thermal Physics

第三章 热力学第一定律

Chapter 3 The First Law of Thermodynamics

一、单项选择题(Single Choice Questions)

1. 一定量的气体在某一过程中，外界对气体做了 $8 \times 10^4 \text{ J}$ 的功，气体的内能减少了 $1.2 \times 10^5 \text{ J}$ ，则下列各式中正确的是()
(In a certain process of a certain amount of gas, the external environment does work of $8 \times 10^4 \text{ J}$ on the gas, and the internal energy of the gas decreases by $1.2 \times 10^5 \text{ J}$. Which of the following expressions is correct?)

- A. $W = 8 \times 10^4 \text{ J}$, $\Delta U = 1.2 \times 10^5 \text{ J}$, $Q = 4 \times 10^4 \text{ J}$
- B. $W = 8 \times 10^4 \text{ J}$, $\Delta U = -1.2 \times 10^5 \text{ J}$, $Q = -2 \times 10^5 \text{ J}$
- C. $W = -8 \times 10^4 \text{ J}$, $\Delta U = 1.2 \times 10^5 \text{ J}$, $Q = 2 \times 10^5 \text{ J}$
- D. $W = -8 \times 10^4 \text{ J}$, $\Delta U = -1.2 \times 10^5 \text{ J}$, $Q = -4 \times 10^4 \text{ J}$

答案(Answer):B

答案解析(中文): 本题考查热力学第一定律的基本应用，高中物理通用符号规则: 外界对气体做功， $W > 0$ ；气体对外界做功， $W < 0$ ；气体吸热， $Q > 0$ ；气体放热， $Q < 0$ ；内能增大， $\Delta U > 0$ ；内能减小， $\Delta U < 0$ 。热力学第一定律的表达式为 $\Delta U = Q + W$ 。已知外界对气体做功 $W = 8 \times 10^4 \text{ J}$ ，气体内能减少故 $\Delta U = -1.2 \times 10^5 \text{ J}$ ，将其代入公式得 $Q = \Delta U - W = -1.2 \times 10^5 - 8 \times 10^4 = -2 \times 10^5 \text{ J}$ ，对应选项 B。

Answer Analysis (English): This question examines the basic application of the first law of thermodynamics. The general sign rules in high school physics are as follows: $W > 0$ when the external environment does work on the gas, $W < 0$ when the gas does work on the external environment; $Q > 0$ when the gas absorbs heat, $Q < 0$ when the gas releases heat; $\Delta U > 0$ when the internal energy increases, $\Delta U < 0$ when the internal energy decreases. The expression of the first law of thermodynamics is $\Delta U = Q + W$. It is known that the external environment does work $W = 8 \times 10^4 \text{ J}$ on the gas, and the internal energy of the gas decreases, so $\Delta U = -1.2 \times 10^5 \text{ J}$. Substitute the values into the formula to get $Q = \Delta U - W = -1.2 \times 10^5 - 8 \times 10^4 = -2 \times 10^5 \text{ J}$, which corresponds to option B.

2. 一定质量的理想气体，从某一状态开始，经过一系列变化后又回到开始的状态，用 W_1 表示外界对气体做的功， W_2 表示气体对外界做的功， Q_1 表示气体吸收的热量， Q_2 表示气体放出的热量，则在整个过程中一定有()
(A certain mass of ideal gas starts from a certain state and returns to the initial state after a series of changes. Let W_1 be the work done by the external environment on the gas, W_2 be the work done by the gas on the surroundings, Q_1 be the heat absorbed by the gas, and Q_2 be the heat released by the gas. Which of the following must be true in the whole process?)

- A. $Q_1 - Q_2 = W_2 - W_1$
- B. $Q_1 = Q_2$
- C. $W_1 = W_2$
- D. $Q_1 > Q_2$

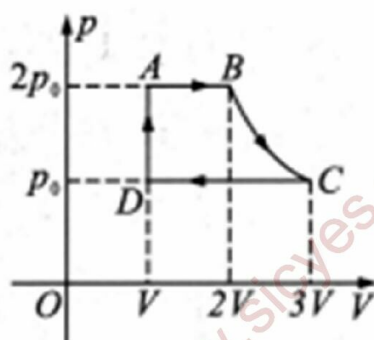


答案(Answer):A

答案解析(中文):理想气体的内能仅由温度决定, 完成循环过程回到初始状态时温度不变, 因此内能变化 $\Delta U = 0$ 。根据热力学第一定律 $\Delta U = Q + W$, 可得总热量与总功的关系为 $Q + W = 0$ 。其中总功 $W = W_1 - W_2$ (外界对气体做功为正, 气体对外做功为负), 总热量 $Q = Q_1 - Q_2$ (气体吸热为正, 放热为负), 代入得 $Q_1 - Q_2 + W_1 - W_2 = 0$, 整理后为 $Q_1 - Q_2 = W_2 - W_1$, 故 A 正确; 循环过程中气体对外与外界对气体的做功大小、吸热与放热的大小均无必然相等关系, B、C 错误; Q_1 和 Q_2 的大小关系由做功情况决定, 并非一定 $Q_1 > Q_2$, D 错误。

Answer Analysis (English):The internal energy of an ideal gas is determined only by temperature. When the gas completes a cyclic process and returns to the initial state, the temperature remains unchanged, so the change in internal energy $\Delta U = 0$. According to the first law of thermodynamics $\Delta U = Q + W$, the relationship between the total heat and the total work is $Q + W = 0$. The total work $W = W_1 - W_2$ ($W > 0$ when the external environment does work on the gas, $W < 0$ when the gas does work on the external environment), and the total heat $Q = Q_1 - Q_2$ ($Q > 0$ when the gas absorbs heat, $Q < 0$ when the gas releases heat). Substitute the values to get $Q_1 - Q_2 + W_1 - W_2 = 0$, which can be rearranged to $Q_1 - Q_2 = W_2 - W_1$, so option A is correct. In a cyclic process, there is no inevitable equality between the work done by the gas on the outside and the work done by the outside on the gas, nor between the heat absorbed and the heat released by the gas, so B and C are wrong; the magnitude relationship between Q_1 and Q_2 is determined by the work done, and it is not necessarily $Q_1 > Q_2$, so D is wrong.

3.一定质量的理想气体从状态 A 开始, 经过 $A \rightarrow B$ 、 $B \rightarrow C$ 、 $C \rightarrow D$ 、 $D \rightarrow A$ 最后回到初始状态 A, 各状态参量如图所示, 则下列说法中正确的是() (A certain mass of ideal gas starts from state A, goes through $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ and finally returns to the initial state A, with all state parameters shown in the figure. Which of the following statements is correct?)



- A. $A \rightarrow B$ 过程气体从外界吸热 (The gas absorbs heat from the outside in the process of $A \rightarrow B$)
- B. $A \rightarrow B$ 过程气体对外做功大于 $C \rightarrow D$ 过程外界对气体做功 The work done by the gas on the outside in the process of $A \rightarrow B$ is greater than the work done by the outside on the gas in the process of $C \rightarrow D$)
- C. $C \rightarrow D$ 过程气体从外界吸热 (The gas absorbs heat from the outside in the process of $C \rightarrow D$)
- D. A 状态气体分子平均动能小于 D 状态气体分子平均动能 (The average kinetic energy of gas molecules in state A is less than that in state D)



答案(Answer):A

答案解析(中文):结合理想气体状态方程 $\frac{PV}{T} = C$ 和热力学第一定律分析, 本题题图为 P-V 图,

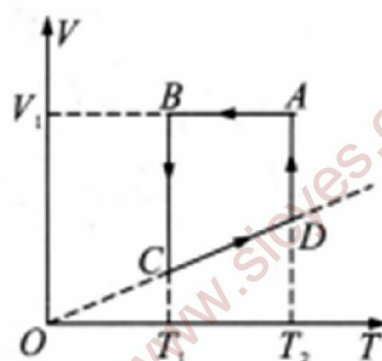
各过程特点为 $A \rightarrow B$ 等压、 $B \rightarrow C$ 等容、 $C \rightarrow D$ 等压、 $D \rightarrow A$ 等容。①A 选项: $A \rightarrow B$ 等压过程, 体积增大则气体温度升高, 理想气体内能增大即 $\Delta U > 0$; 体积增大气体对外做功 $W < 0$, 由 $\Delta U = Q + W$ 得 $Q > 0$, 气体从外界吸热, A 正确。②B 选项:等压过程气体做功大小由 $|W| = p|\Delta V|$ 计算(做功的绝对值), 由题图可知 $A \rightarrow B$ 和 $C \rightarrow D$ 过程的 $p|\Delta V|$ 相等, 故做功大小相等, B 错误。③C 选项: $C \rightarrow D$ 等压过程, 体积减小则温度降低, $\Delta U < 0$; 外界对气体做功 $W > 0$, 由 $\Delta U = Q + W$ 得 $Q < 0$, 气体向外界放热, C 错误。④D 选项:分子平均动能仅由温度决定, 由理想气体状态方程可知 A 状态温度高于 D 状态, 故 A 状态气体分子平均动能更大, D 错误。

Answer Analysis (English):Analyze combined with the ideal gas state equation $\frac{PV}{T} = C$ and

the first law of thermodynamics. The figure of this question is a P-V diagram, and the characteristics of each process are isobaric for $A \rightarrow B$, isochoric for $B \rightarrow C$, isobaric for $C \rightarrow D$, and isochoric for $D \rightarrow A$. ①Option A: In the isobaric process of $A \rightarrow B$, the volume increases so the gas temperature rises, and the internal energy of the ideal gas increases, that is $\Delta U > 0$; the gas does work on the outside with $W < 0$ due to the increased volume. From $\Delta U = Q + W$, we get $Q > 0$, meaning the gas absorbs heat from the outside, so A is correct. ②Option B: The magnitude of work done by the gas in an isobaric process is calculated by $|W| = p|\Delta V|$ (the absolute value of work done). It can be seen from the figure that $p|\Delta V|$ in the processes of $A \rightarrow B$ and $C \rightarrow D$ are equal, so the work done is equal in magnitude, so B is wrong. ③Option C: In the isobaric process of $C \rightarrow D$, the volume decreases so the temperature drops, and $\Delta U < 0$; the external environment does work on the gas with $W > 0$. From $\Delta U = Q + W$, we get $Q < 0$, meaning the gas releases heat to the outside, so C is wrong. ④Option D: The average kinetic energy of molecules is determined only by temperature. It can be known from the ideal gas state equation that the temperature of state A is higher than that of state D, so the average kinetic energy of gas molecules in state A is larger, so D is wrong.



4. 某汽缸内封闭有一定质量的理想气体, 从状态 A 依次经过状态 B、C 和 D 后再回到状态 A, 其 $V-T$ 图像如图所示, 则在该循环过程中, 下列说法中正确的是() (A certain mass of ideal gas is enclosed in a cylinder, which goes through state B, C and D in turn from state A and then returns to state A, and its $V-T$ graph is shown in the figure. Which of the following statements is correct in this cyclic process?)



- A. 从 B 到 C 过程中, 气体吸收热量 (The gas absorbs heat in the process from B to C)
- B. 从 C 到 D 过程中, 气体的压强增大 (The pressure of the gas increases in the process from C to D)
- C. 从 D 到 A 过程中, 单位时间内碰撞器壁单位面积的分子数减少 (The number of molecules colliding with unit area of the container wall per unit time decreases in the process from D to A)
- D. 若气体在从 C 到 D 过程中, 内能增加 3 kJ, 对外做功 5 kJ, 则气体向外界放出热量 8 kJ (If the internal energy of the gas increases by 3 kJ and the gas does work of 5 kJ on the outside in the process from C to D, the gas releases heat of 8 kJ to the outside)

答案(Answer): C

答案解析(中文): 结合 $V-T$ 图像的过程特点和理想气体状态方程、热力学第一定律分析:

① **A 选项:** B→C 为等容过程 ($V-T$ 图像中体积不变), 气体不做功即 $W=0$; 温度降低则理想气体内能减小, $\Delta U < 0$, 由 $\Delta U = Q + W$ 得 $Q < 0$, 气体放出热量, A 错误。

② **B 选项:** 由理想气体状态方程 $\frac{PV}{T} = C$ 变形得 $V = \frac{C}{P}T$ 。C→D 过程温度 T 不变、体积 V 增大, 故压强 P 减小, B 错误。

③ **C 选项:** D→A 过程的 $V-T$ 图像为过原点的直线, 满足 $V/T = \text{常数}$, 由理想气体状态方程可知是等压过程, 压强不变; 温度升高, 分子平均动能增大, 而压强由分子数密度和平均动能共同决定, 压强不变、平均动能增大, 说明分子数密度减小 (体积增大), 因此单位时间内碰撞器壁单位面积的分子数减少, C 正确。

④ **D 选项:** 已知 $\Delta U = 3 \text{ kJ}$, 气体对外做功 $W = -5 \text{ kJ}$, 代入 $\Delta U = Q + W$ 得 $Q = \Delta U - W = 3 - (-5) = 8 \text{ kJ}$, $Q > 0$ 说明气体从外界吸收 8 kJ 热量, 而非放出, D 错误。

Answer Analysis (English): Analyze combined with the process characteristics of the $V-T$ graph, the ideal gas state equation and the first law of thermodynamics:

① **Option A:** B→C is an isochoric process (the volume remains unchanged in the $V-T$ graph), and the gas does no work, that is $W=0$; the temperature decreases so the internal energy of the ideal gas decreases, $\Delta U < 0$. From $\Delta U = Q + W$, we get $Q < 0$, meaning



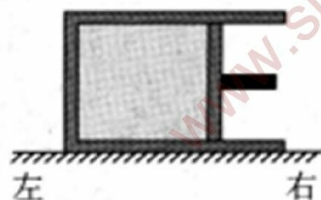
the gas releases heat, so A is wrong.

② **Option B:** Rearrange the ideal gas state equation $\frac{PV}{T} = C$ to $V = \frac{C}{P}T$. In the process of $C \rightarrow D$, the temperature T is constant and the volume V increases, so the pressure P decreases, so B is wrong.

③ **Option C:** The $V - T$ graph of the $D \rightarrow A$ process is a straight line passing through the origin, satisfying $V/T = \text{constant}$. According to the ideal gas state equation, this is an isobaric process with constant pressure. The temperature rises, so the average kinetic energy of molecules increases. Since pressure is determined by both number density of molecules and average kinetic energy, and the pressure remains constant while average kinetic energy increases, it indicates that the number density of molecules decreases (volume increases). Therefore, the number of molecules colliding with unit area of the container wall per unit time decreases, so C is correct.

④ **Option D:** It is known that $\Delta U = 3 \text{ kJ}$, and the gas does work on the outside with $W = -5 \text{ kJ}$. Substitute into $\Delta U = Q + W$ to get $Q = \Delta U - W = 3 - (-5) = 8 \text{ kJ}$. $Q > 0$ indicates that the gas absorbs 8 kJ of heat from the outside, not releases, so D is wrong.

5. 如图所示, 固定在水平面上的汽缸内封闭着一定质量的理想气体, 汽缸壁和活塞绝热性能良好, 汽缸内气体分子间相互作用的势能忽略不计, 则下列说法中正确的是() (As shown in the figure, a certain mass of ideal gas is enclosed in a cylinder fixed on the horizontal plane. The cylinder wall and the piston are well insulated, and the potential energy of the interaction between gas molecules in the cylinder is negligible. Which of the following statements is correct?)



- A. 使活塞向左移动, 汽缸内气体对外界做功, 内能减少 (Move the piston to the left, the gas in the cylinder does work on the outside and its internal energy decreases)
- B. 使活塞向左移动, 汽缸内气体内能增大, 温度升高 (Move the piston to the left, the internal energy of the gas in the cylinder increases and its temperature rises)
- C. 使活塞向左移动, 汽缸内气体压强减小 (Move the piston to the left, the pressure of the gas in the cylinder decreases)
- D. 使活塞向左移动, 汽缸内气体分子无规则运动的平均动能减小 (Move the piston to the left, the average kinetic energy of the random motion of gas molecules in the cylinder decreases)

答案(Answer):B

答案解析(中文): 汽缸壁和活塞绝热性能良好, 说明气体与外界无热交换, 即 $Q = 0$; 理想气体分子间势能忽略不计, 其内能仅由温度决定, 分子无规则运动的平均动能也仅由温度决定。活塞向左移动时, 气体体积减小, 外界对气体做功, $W > 0$ 。根据热力学第一定律 $\Delta U = Q + W$, 得 $\Delta U > 0$, 气体内能增大, 温度升高, 分子无规则运动的平均动能增大, 故 B

正确, A、D 错误; 由理想气体状态方程 $\frac{PV}{T} = C$, 体积 V 减小、温度 T 升高, 故压强 P



增大，C 错误。

Answer Analysis (English): The cylinder wall and the piston are well insulated, which means there is no heat exchange between the gas and the surroundings, that is $Q = 0$; the potential energy of the interaction between ideal gas molecules is negligible, and its internal energy is determined only by temperature, as is the average kinetic energy of molecular random motion. When the piston moves to the left, the gas volume decreases, and the external environment does work on the gas with $W > 0$. According to the first law of thermodynamics $\Delta U = Q + W$, we get $\Delta U > 0$, the internal energy of the gas increases, the temperature rises, and the average kinetic energy of molecular random motion increases, so B is correct, A and D are wrong; from the ideal gas state equation $\frac{PV}{T} = C$, the volume V decreases and the temperature T rises, so the pressure P increases, so C is wrong.



6. 在商场一些柜台前常见到一种气压凳，如图所示，它的高度可以调节且舒适性好，其原理结构简化模型就是一个汽缸活塞模型，内部封闭一定质量氮气。如果有一个人坐上去，关于内部气体(可以看作理想气体，忽略与外界热交换)的说法中正确的是() (As shown in the figure, a pneumatic stool is commonly seen in front of some counters in shopping malls, which has an adjustable height and good comfort. Its simplified structural model is a cylinder-piston model with a certain mass of nitrogen gas enclosed inside. If a person sits on it, which of the following statements about the internal gas (regarded as an ideal gas, neglecting heat exchange with the surroundings) is correct?)

- A. 内能减小(The internal energy decreases)
- B. 温度不变(The temperature remains unchanged)
- C. 压强变大(The pressure increases)
- D. 每个气体分子的速率都变大(The speed of each gas molecule increases)

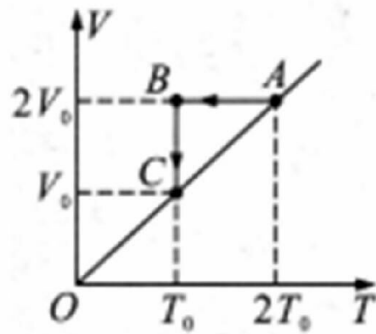
答案(Answer): C

答案解析(中文): 忽略气体与外界的热交换，即 $Q = 0$ ；人坐上去后活塞向下移动，外界对封闭气体做功， $W > 0$ 。①A、B 选项：由热力学第一定律 $\Delta U = Q + W$ 得 $\Delta U > 0$ ，理想气体内能增大，温度升高，A、B 错误。②C 选项：由理想气体状态方程 $\frac{PV}{T} = C$ ，气体体积 V 减小、温度 T 升高，故压强 P 一定增大，C 正确。③D 选项：温度升高表示气体分子的平均速率增大，这是统计规律，仅反映整体分子的运动情况，并非每个气体分子的速率都变大，部分分子速率可能减小、不变，D 错误。

Answer Analysis (English): Neglect heat exchange between the gas and the surroundings, that is $Q = 0$; when a person sits on the stool, the piston moves down, and the external environment does work on the enclosed gas with $W > 0$. ①Options A and B: From the first law of thermodynamics $\Delta U = Q + W$, we get $\Delta U > 0$, the internal energy of the ideal



gas increases and the temperature rises, so A and B are wrong. ②Option C: From the ideal gas state equation $\frac{PV}{T} = C$, the gas volume V decreases and the temperature T rises, so the pressure P must increase, so C is correct. ③Option D: A rise in temperature means the average speed of gas molecules increases, which is a statistical law and only reflects the motion of the whole molecules. Not every gas molecule's speed increases; the speed of some molecules may decrease or remain unchanged, so D is wrong.



7.如图所示,为一定质量的氦气(可视为理想气体)状态变化的 $V-T$ 图像。已知该氦气所含的氦分子总数为 N , 氦气的摩尔质量为 M , 在状态 A 时的压强为 p_0 , 阿伏加德罗常数为 N_A 。下列说法中正确的是()
(As shown in the figure, it is the $V-T$ graph of the state change of a certain mass of helium gas (the original question has a writing error of ammonia/nitrogen gas, regarded as an ideal gas). It is known that the total number of helium molecules in the helium gas is N , the molar mass of helium gas is M , the pressure at state A is p_0 , and the Avogadro constant is N_A . Which of the following statements is correct?)

- A. 氦气分子的质量为 $\frac{M}{N}$ (The mass of a helium molecule is M/N)
- B. 状态 B 时氦气的压强为 $2p_0$ (The pressure of helium gas at state B is $2p_0$)
- C. $B \rightarrow C$ 过程中氦气向外界放热 (Helium gas releases heat to the outside in the process of $B \rightarrow C$)
- D. 状态 C 时氦气分子间的平均距离 $d = \sqrt[3]{\frac{V_0}{N_A}}$ (The average distance between helium molecules at state C is $d = \sqrt[3]{\frac{V_0}{N_A}}$)

答案(Answer):C

答案解析(中文):结合分子质量计算、 $V-T$ 图像的过程特点、热力学第一定律和分子平均距离计算分析。①A 选项:分子质量的计算公式为 $m = \frac{\text{摩尔质量}}{\text{阿伏加德罗常数}}$, 因此氦气分子的质量

为 $m = \frac{M}{N_A}$, 而非 $\frac{M}{N}$, A 错误。②B 选项: $V-T$ 图像中过坐标原点的直线为等压过程, A、

B 在同一直线上, 故 $A \rightarrow B$ 为等压过程, 状态 B 的压强 $p_B = p_0$, B 错误。③C 选项: $B \rightarrow C$ 为等容过程, 气体不做功即 $W = 0$; 温度降低则理想气体内能减小, $\Delta U < 0$, 由 $\Delta U = Q + W$ 得 $Q < 0$, 氦气向外界放热, C 正确。④D 选项:设状态 C 时气体体积为 V_0 , 分子总数为 N , 将气体分子占据的空间视为立方体, 分子间平均距离为立方体的边长, 即 $d =$



$\sqrt[3]{\frac{V_0}{N}}$, 而非 $\sqrt[3]{\frac{V_0}{N_A}}$, D 错误。

Answer Analysis (English):Analyze combined with molecular mass calculation, process characteristics of the V-T graph, the first law of thermodynamics and average molecular

distance calculation. ①Option A: The formula for molecular mass is $m = \frac{\text{Molar Mass}}{\text{Avogadro Constant}}$, so

the mass of a helium molecule is $m = \frac{M}{N_A}$, not $\frac{M}{N}$, so A is wrong. ②Option B: The straight

line passing through the origin in the V-T graph is an isobaric process. A and B are on the same straight line, so $A \rightarrow B$ is an isobaric process, and the pressure of state B is $p_B = p_0$,

so B is wrong. ③Option C: $B \rightarrow C$ is an isochoric process, and the gas does no work, that is $W = 0$; the temperature drops so the internal energy of the ideal gas decreases, $\Delta U < 0$.

From $\Delta U = Q + W$, we get $Q < 0$, meaning helium gas releases heat to the outside, so C

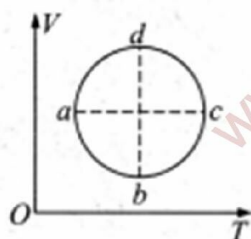
is correct. ④Option D: Let the gas volume at state C be V_0 and the total number of molecules be N . Regard the space occupied by gas molecules as a cube, and the average

distance between molecules is the side length of the cube, that is $d = \sqrt[3]{\frac{V_0}{N}}$, not $\sqrt[3]{\frac{V_0}{N_A}}$, so D

is wrong.

8.一定质量理想气体的状态变化如图所示, 该气体从状态 a 沿圆形线到状态 b、c、d, 最终

回到状态 a, 则() (As shown in the figure, it is the state change of a certain mass of ideal gas. The gas goes from state a to state b, c and d along a circular line and finally returns to state a. Which of the following statements is correct?)



- A. 状态 b 到状态 d 的过程是等温变化过程 (The process from state b to state d is an isothermal change process)
- B. 状态 a 的压强大于状态 b 的压强 (The pressure of state a is greater than that of state b)
- C. 从状态 c 到状态 d, 体积减小 (The volume decreases in the process from state c to state d)
- D. 从状态 a 到状态 c, 内能增大 (The internal energy increases in the process from state a to state c)

答案(Answer):D

答案解析(中文):本题题图为 V-T 图像(纵坐标为体积 V , 横坐标为温度 T), 结合理想气体状态方程 $PV = nRT$ 和内能的决定因素分析。

① **A 选项:**理想气体等温变化的 V-T 图像为竖直线(温度 T 不变), 而从状态 b 到 d 的过程中, 温度 T 发生变化(b 点温度低于 d 点温度), 因此该过程不是等温变化, A 错误。

② **B 选项:**由理想气体状态方程变形得 $P = \frac{nRT}{V}$ 。状态 a: $T_a = T_1$, $V_a = V_0$; 状态 b: $T_b =$



T_0 , $V_b = V_0 - \Delta V$ (其中 $T_1 < T_0$)。代入得 $P_a = \frac{nRT_1}{V_0}$, $P_b = \frac{nR_0}{V_0 - \Delta V}$ 。由于 $T_1 < T_0$ 且 $V_0 > V_0 - \Delta V$, 可得 $P_a < P_b$, 因此 B 错误。

③ **C 选项**: V-T 图像的纵坐标为体积 V , 从状态 c 到 d 的过程中, 纵坐标数值增大, 说明气体体积增大, C 错误。

④ **D 选项**: 理想气体的内能仅由温度决定。从状态 a 到 c 的过程中, 温度 T 升高($T_c > T_a$), 因此内能增大, D 正确。

Answer Analysis (English): The figure of this question is a **V-T diagram** (ordinate is volume V , abscissa is temperature T), analyzed combined with the ideal gas state equation $PV = nRT$ and the determinant of internal energy.

① **Option A**: The V-T graph of the isothermal change of an ideal gas is a vertical line (temperature T is constant). However, during the process from state b to d, the temperature T changes (the temperature at point b is lower than that at point d), so this process is not an isothermal change. Thus, A is wrong.

② **Option B**: Rearrange the ideal gas state equation to $P = \frac{nRT}{V}$. For state a: $T_a = T_1$, $V_a = V_0$; for state b: $T_b = T_0$, $V_b = V_0 - \Delta V$ (where $T_1 < T_0$). Substituting gives $P_a = \frac{nRT_1}{V_0}$ and $P_b = \frac{nRT_0}{V_0 - \Delta V}$. Since $T_1 < T_0$ and $V_0 > V_0 - \Delta V$, we can conclude $P_a < P_b$. Thus, B is wrong.

③ **Option C**: The ordinate of the V-T diagram is the volume V . During the process from state c to d, the ordinate value increases, indicating that the gas volume increases. Thus, C is wrong.

④ **Option D**: The internal energy of an ideal gas is determined only by temperature. During the process from state a to c, the temperature T increases ($T_c > T_a$), so the internal energy increases. Thus, D is correct.



9. 有一种在超市中常见的“强力吸盘挂钩”，如图甲所示，图乙、图丙是其工作原理示意图。使用时，按住锁扣把吸盘紧压在墙上（如图乙），然后把锁扣扳下（如图丙），让锁扣以盘盖为依托把吸盘向外拉出，使吸盘牢牢地被固定在墙壁上。若吸盘内气体可视为理想气体，且温度始终保持不变，则此过程中() (As shown in Figure 甲, 乙 and 丙, there is a kind of “strong suction cup hook” commonly seen in supermarkets, and Figure 乙 and 丙 are schematic diagrams of its working principle. When in use, press the lock and press the suction cup tightly against the wall, then pull the lock down and pull the suction cup outwards with the cover as the support, so that the suction cup is firmly fixed on the wall. If the gas inside the suction cup is regarded as an ideal gas and its temperature remains constant all the time, which of the following is true in this process?)



- A. 吸盘内气体压强增大(The pressure of the gas inside the suction cup increases)
- B. 吸盘内气体分子的密度增大(The density of gas molecules inside the suction cup increases)
- C. 吸盘内气体分子的平均速率增大(The average speed of gas molecules inside the suction cup increases)
- D. 吸盘内气体要吸收热量(The gas inside the suction cup absorbs heat from the outside)

答案(Answer):D

答案解析(中文):吸盘内气体的温度始终不变，为等温过程，理想气体等温过程遵循玻意耳定律 $PV = C$ (恒量)，结合热力学第一定律和分子运动的统计规律分析。①A选项：扳下锁扣拉出吸盘，吸盘内气体体积 V 增大，由玻意耳定律 $PV = C$ 得气体压强 P 减小，A 错误。②B选项：气体分子总数不变，体积增大则分子数密度(单位体积内的分子数)减小，B 错误。③C选项：分子的平均速率仅由温度决定，温度不变则分子平均速率不变，C 错误。④D选项：等温过程中理想气体的内能不变，即 $\Delta U = 0$ ；体积增大，气体对外界做功， $W < 0$ ，由热力学第一定律 $\Delta U = Q + W$ 得 $Q > 0$ ，说明吸盘内气体从外界吸收热量，D 正确。

Answer Analysis (English):The temperature of the gas inside the suction cup remains constant all the time, which is an isothermal process. The isothermal process of an ideal gas follows Boyle's law $PV = C$ (constant). Analyze combined with the first law of thermodynamics and the statistical law of molecular motion. ①Option A: Pull down the lock and pull out the suction cup, the volume V of the gas inside the suction cup increases. From Boyle's law $PV = C$, the gas pressure P decreases, so A is wrong. ②Option B: The total number of gas molecules is constant, and the number density of molecules (the number of molecules per unit volume) decreases as the volume increases, so B is wrong. ③Option C: The average speed of molecules is determined only by temperature, and the average speed of molecules remains constant when the temperature is constant, so C is wrong. ④Option D: The internal energy of the ideal gas remains unchanged in the isothermal process, that is $\Delta U = 0$; the volume increases, and the gas does work on the outside with $W < 0$. From the first law of thermodynamics $\Delta U = Q + W$, we get $Q > 0$, indicating that the gas inside



the suction cup absorbs heat from the outside, so D is correct.



10.如图所示，为某同学设计的一个简易温度计，一根透明吸管插入导热良好的容器，连接处密封，在吸管内注入一小段油柱，外界大气压保持不变。将容器放入热水中，观察到油柱缓慢上升，下列说法中正确的是() (As shown in the figure, it is a simple thermometer designed by a student. A transparent straw is inserted into a container with good thermal conductivity, the connection is sealed, a short oil column is injected into the straw, and the external atmospheric pressure remains constant. When the container is put into hot water, the oil column is observed to rise slowly. Which of the following statements is correct?)

- A. 气体对外做的功大于吸收的热量(The work done by the gas on the outside is greater than the heat absorbed)
- B. 气体对外做的功等于吸收的热量(The work done by the gas on the outside is equal to the heat absorbed)
- C. 容器内壁的单位面积上受到气体分子的平均作用力增大(The average force exerted by gas molecules on the unit area of the container inner wall increases)
- D. 容器内壁的单位面积上受到气体分子的平均作用力大小不变(The magnitude of the average force exerted by gas molecules on the unit area of the container inner wall remains unchanged)

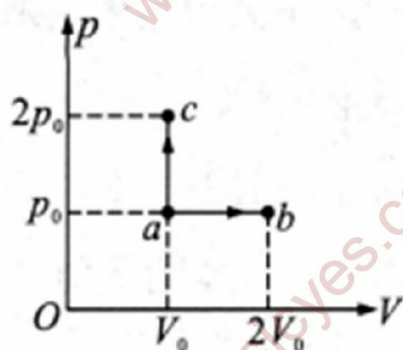
答案(Answer):D

答案解析(中文):容器导热性能良好，气体与热水发生热传递；外界大气压保持不变，气体做等压变化，结合热力学第一定律和压强的微观意义分析。①A、B选项：油柱上升说明气体体积增大，气体对外界做功， $W < 0$ ；容器放入热水中，气体温度升高，理想气体内能增大， $\Delta U > 0$ 。由热力学第一定律 $\Delta U = Q + W$ 得 $Q = \Delta U - W$ ，因此 $Q > |W|$ ，即气体吸收的热量大于对外做的功，A、B 错误。②C、D 选项：容器内壁单位面积上受到气体分子的平均作用力等于气体的压强，本题中气体为等压变化，压强始终等于外界大气压，故该平均作用力大小不变，C 错误，D 正确。

Answer Analysis (English):The container has good thermal conductivity, and the gas exchanges heat with hot water; the external atmospheric pressure remains constant, and the gas undergoes isobaric change. Analyze combined with the first law of thermodynamics and the microscopic meaning of pressure. ①Options A and B: The rise of the oil column indicates that the gas volume increases, and the gas does work on the outside with $W < 0$; the container is put into hot water, the gas temperature rises, and the internal energy of the ideal gas increases, $\Delta U > 0$. From the first law of thermodynamics $\Delta U = Q + W$, we get $Q = \Delta U - W$, so $Q > |W|$, meaning the heat absorbed by the gas is greater than the work done on the outside, so A and B are wrong. ②Options C and D: The average force exerted by gas molecules on the unit area of the container inner wall is equal to the gas pressure. In



this question, the gas undergoes isobaric change, and the pressure is always equal to the external atmospheric pressure, so the magnitude of the average force remains unchanged, so C is wrong and D is correct.



11.如图所示，一定质量的理想气体，由状态 a 经过 ab 过程到达状态 b 或者经过 ac 过程到达状态 c 。设气体在状态 b 和状态 c 的内能分别为 U_b 和 U_c ，在过程 ab 和 ac 中吸收的热量分别为 Q_{ab} 和 Q_{ac} 。则() (As shown in the figure, a certain mass of ideal gas reaches state b from state a through the ab process or state c through the ac process. Let the internal energy of the gas at state b and state c be U_b and U_c respectively, and the heat absorbed in the ab and ac processes be Q_{ab} and Q_{ac} respectively. Which of the following is true?)

- A. $U_b > U_c$, $Q_{ab} > Q_{ac}$
- B. $U_b > U_c$, $Q_{ab} < Q_{ac}$
- C. $U_b = U_c$, $Q_{ab} > Q_{ac}$
- D. $U_b = U_c$, $Q_{ab} < Q_{ac}$

答案(Answer):C

答案解析(中文):本题核心考查理想气体内能的决定因素和不同过程的做功分析，由题图可知 b 、 c 两点在同一条等温线上。①内能关系:理想气体的内能仅由温度决定， b 、 c 两点温度相同，因此 $U_b = U_c$ ，A、B 错误。②吸热关系: a 到 b 和 a 到 c 过程的内能变化 ΔU 相同。 $a \rightarrow b$ 过程气体体积增大，气体对外界做功， $W_{ab} < 0$ ； $a \rightarrow c$ 过程为等容过程，气体不做功， $W_{ac} = 0$ 。根据热力学第一定律 $\Delta U = Q + W$ ，得 $Q_{ab} = \Delta U - W_{ab}$ ， $Q_{ac} = \Delta U - W_{ac} = \Delta U$ 。由于 $W_{ab} < 0$ ，故 $Q_{ab} > \Delta U = Q_{ac}$ ，C 正确，D 错误。

Answer Analysis (English):This question mainly examines the determinant of the internal energy of an ideal gas and the work analysis of different processes. It can be seen from the figure that points b and c are on the same isothermal line. ①Internal energy relationship: The internal energy of an ideal gas is determined only by temperature. The temperatures of points b and c are the same, so $U_b = U_c$ ，so A and B are wrong. ②Heat absorption relationship: The change in internal energy ΔU of the processes from a to b and a to c is the same. In the process of $a \rightarrow b$ ，the gas volume increases, and the gas does work on the outside with $W_{ab} < 0$ ；the process of $a \rightarrow c$ is an isochoric process, and the gas does no work with $W_{ac} = 0$ 。According to the first law of thermodynamics $\Delta U = Q + W$ ，we get $Q_{ab} = \Delta U - W_{ab}$ and $Q_{ac} = \Delta U - W_{ac} = \Delta U$ 。Since $W_{ab} < 0$ ， $Q_{ab} > \Delta U = Q_{ac}$ ，so C is correct and D is wrong.